

Precise Elbow Angle Measurement for Exoskeletons Applications: Comparison between Optical Fiber and IMU-based sensors

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Abstract: The accuracy of measuring elbow position is essential in human motion analysis and control strategies for robotic exoskeletons. Currently, the technology for measuring position includes Inertial Measurement Units (IMU), encoders, and optical fiber (OF). However, these methods suffer from integration errors or are limited to rigid structures. Optical fiber has proven to be a promising solution for measuring elbow position, being flexible, low cost, and immune to electromagnetic interference. This article will compare OF and IMU sensors to determine the suitable option for measuring elbow position in exoskeleton applications. Five healthy subjects performed a repetitive lifting activity using the IMU, OF, and encoder. These results indicate a low difference between the OF and the encoder, making the OF a suitable solution due to cost, precision, and ease of implementation. In conclusion, when comparing these two methods, the OF sensor has multiple advantages over commercial sensors to be implemented in exoskeleton applications.

Keywords Elbow, angle sensor, optical fiber, robotics

1. Introduction

The upper limb performs many activities of daily living, which leads to a significant impact on people's quality of life [13]. Specifically, the elbow's joint serves as a mechanical connection between the hand, wrist, and shoulder. Its functions include positioning the hand in space for precise movements, powerful grasping, and other multiple activities [6]. Motion estimation and tracking of continuous joint movement of human arms are essential to human-robot interaction [8]. Therefore, it is important to monitor and evaluate the movement and function of the elbow joint to provide information about the person's degree of physical activity and functional capacity [5].

Precise estimation of the elbow's angle is essential in robotic applications where biomechanical methods are utilized to estimate the elbow's range of motion (ROM) and angular displacements [2]. The monitoring of this parameter is used in the control strategies of robotic exoskeletons for motion assistance, and rehabilitation for activities of daily living [10].

Currently, technology for measuring position for evaluating and analyzing exoskeletons includes magnetic encoders, Inertial Measurement Units (IMUs), and optical fiber (OF). IMUs are widely used in industry, but they suffer from an integration error that increases with time and velocity [12]. On the

other hand, magnetic encoders are precise but limited to rigid structures and cannot be used in intense electromagnetic environments [1].

Additionally, OF technology has proven to be a promising solution for measuring elbow position for exoskeleton applications. OF is a flexible structure, low cost, and immune to electromagnetic interference, making it an alternative for mounting on an elbow exoskeleton and measuring parameters [4]. However, a comparison with commercial sensors should be conducted to observe their errors, advantages, and disadvantages. This will help to determine if the OF sensor is the most suitable option for robotic applications.

In this article, it is compared an OF and IMU sensors previously mentioned with the magnetic encoder in terms of accuracy, reliability, and cost to determine which is the most suitable sensor for measuring elbow position for exoskeleton applications.

2. Method

2.1 Materials

A three-dimensional polylactic acid (PLA) structure was constructed for the elbow joint. The structure was secured using velcro, and an OF was threaded through the system. The polymer OF (SH4001, Mitsubishi Chemical Co.) is used to measure the angle of the elbow. A sensitive zone [9] was created where voltage changes would occur in response to the joint's flexion and extension. Also, a Light-emitting diode (LED) IF E96R, and a phototransistor IF-D92 (Industrial Fiber Optics, USA) located on opposite sides of the fiber are utilized to measure changes in voltage. Finally, a microcontroller Teensy 3.6 (PJRC, USA) with an ADC of 16 bits of resolution is used to measure the voltage of the phototransistor and the sensors.

Additionally, an Alt-IMU V4 inertial sensor was positioned on the participant's arm to compare this sensor in measuring the elbow's angle. Finally, a magnetic encoder AS5600 was located at the elbow joint on the rigid structure to provide a standard angle measure. This encoder is used to compare the OF and IMU signals in measuring the joint's angle because of its ability to accurately measure the angle in real-time, providing valuable information for the analysis of the results. Table 1 shows a comparison between the characteristics of the three sensors, and Figure 1 shows the setup including all of the sensors used in the study.

Table 1. Characteristics of the encoder, OF and IMU sensors.

Parameter/Sensor	Encoder	Fiber	IMU
Precision	+/- 0.33°	Depends on each sensitive zone	+/- 2 °/s (Gyr)
Resolution	16-bit (DAC resolution)	16-bit (Teensy 3.6)	16-bit reading per axis (Acc & Gyr)
Mechanic characteristics	Robust environmental tolerance	Flexible and resistant to deformation	Vibration and shock resistant
Acquisition rate	Up to 10 kHz	Up to 10 kHz	Up to 200 Hz
Material	Magnetic and metallic components	PMMA	Electronic components

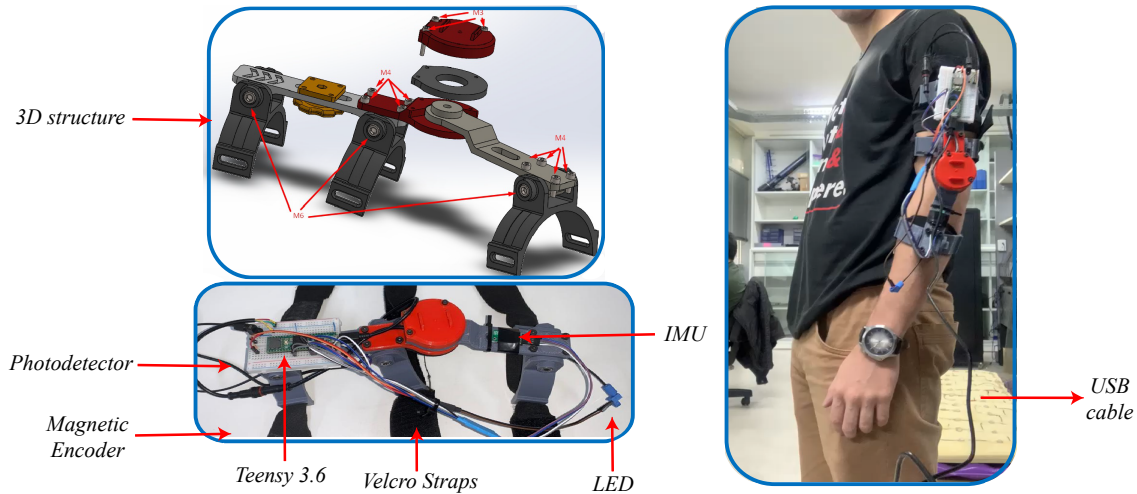


Figure 1. Setup of 3D structure, sensors and participant during the test

2.2 Optical Fiber Characterization

For the OF characterization, a participant performed various movements during arm flexion and extension. This information was acquired and compared with the magnetic encoder data to identify the angles that the person was performing. To generate the sensitive zone of the OF, the jacket of the fiber was removed from the fiber in the elbow flexion area. For this purpose, a razor blade was used where the jacket was cylindrically removed in the corresponding area to the arm flexion/extension. In addition, to obtain greater voltage loss and, therefore, greater sensitivity, a DC motor was used to rotate the fiber to remove the cladding by means of sandpaper slightly. This type of cut in the sensitive zone has been previously evaluated, regulating sensitivity and improving misalignment issues compared to the OF curvature sensors [3], and also evaluating its fatigue or number of cycles that it can withstand [7].

2.3 Experimental Protocol

The protocol is divided into two main stages in five volunteer healthy subjects. All subjects signed the inform consent after reading the protocol approved by the ethics comitte with number 4.635.835. The first stage is a calibration phase, where the subject must stand still for 10 seconds. This allows the sensor to settle and record a stable baseline measurement, which can then be used to subtract the offset from subsequent readings. In the second stage, subjects performed the biceps curl test for multiple repetitions during 2 minutes. This test aims to evaluate the sensor's signals in the cyclic periods of flexion/extension of the elbow.

2.4 Data processing and feature extraction

MATLAB software was used for data processing. First, all three signals were acquired at a sampling frequency of 52 Hz; then, a low-pass filter with a cutoff frequency of 0.5 Hz was applied to reduce noise. A Fast Fourier Transform (FFT) was applied to observe the principal components of the signal. The encoder data were acquired by converting the values from raw encoder counts to degrees. Lastly, to calculate the position of the IMU, the numerical integration method known as the Euler method was used. The method is one of the most used in the literature in low-cost IMU sensors [11]. It also assumes a constant gyroscope rotation rate, and the position accuracy depends on the sampling frequency and the accuracy of the gyroscope. With this processing, variables such as Root Mean Squared (RMS), mean, minimum (MIN), and maximum (MAX) variables of the IMU

and fiber-optic were acquired for the three sensors. These values are calculated per repetition window and averaged per subject.

2.5 Statistical Analysis

The SPSS software was used for the statistical analysis of the three sensors. First, the normality test is performed through the Shapiro-Wilk test. Afterward, the Friedman test is performed by comparing the fiber, encoder, and IMU sensors in all variables to determine any statistically significant differences. Lastly, the U-Mann Whitney test is performed to observe significant changes between the encoder (reference) and the fiber and IMU sensors.

2.6 Results and discussion

The response of the sensor characterization is shown in Figure 2, which demonstrated a linear relationship and an R squared value higher than 95%. This indicated a strong correlation between voltage and angle. Two separate equations were established for the flexion and extension movements due to the flexible nature of the fiber, which resulted in some hysteresis. Table 2 presents the results between

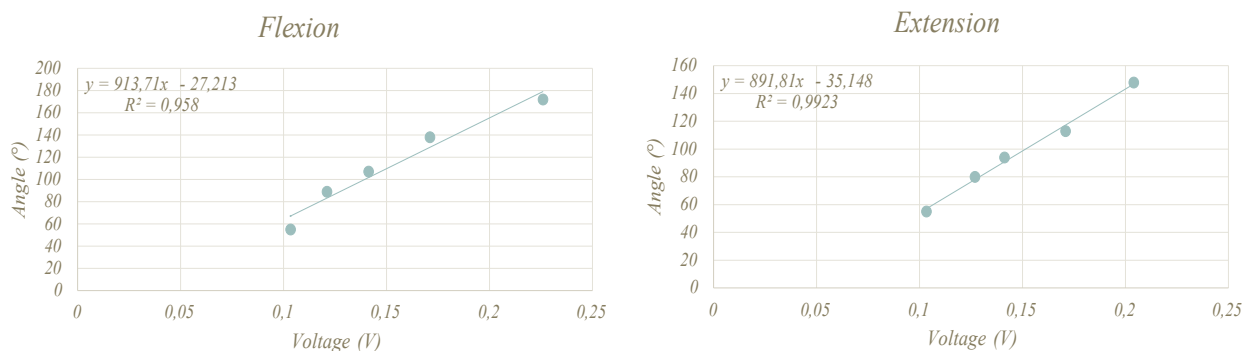


Figure 2. Optical fiber characterization for flexion and extension of the elbow

three sensors: OF, magnetic encoder, and IMU. Five parameters were compared: ROM, RMS, mean, min, and max. Also, the percentage errors when compared with the encoder are presented.

The precision and performance of each sensor are observed with the values obtained for each parameter. For instance, The IMU shows a significant difference in all parameters compared to the encoder, with errors between 6% - 73% indicating that the IMU is less precise in measuring the elbow's angle. This is mainly due to the integration error when the position is calculated. This error causes a drift in the signal, and they accumulate over time due to imperfect data [12]. Besides, additional knowledge, such as initial conditions, is required to determine the integration constants, which leads to the unsuccessful identification of this parameter [11].

On the other hand, the OF and magnetic encoder do not change significantly in the RMS and mean values, suggesting a similarity in their precision and performance. However, values such as ROM, MIN, and MAX with errors between 2% - 14%. This can also be observed in Figure 3, where the encoder and OF have similar graphs. The IMU signal varies due to the errors presented in the integration method. The error is also observed when reaching the minimum or maximum value due to the fiber's flexibility since it still presents hysteresis. This should be considered for adjustment in future work during fiber characterization.

Additionally, this data highlights the importance of considering the limitations and strengths of each sensor when selecting a measurement device for a specific application. It is important to ob-

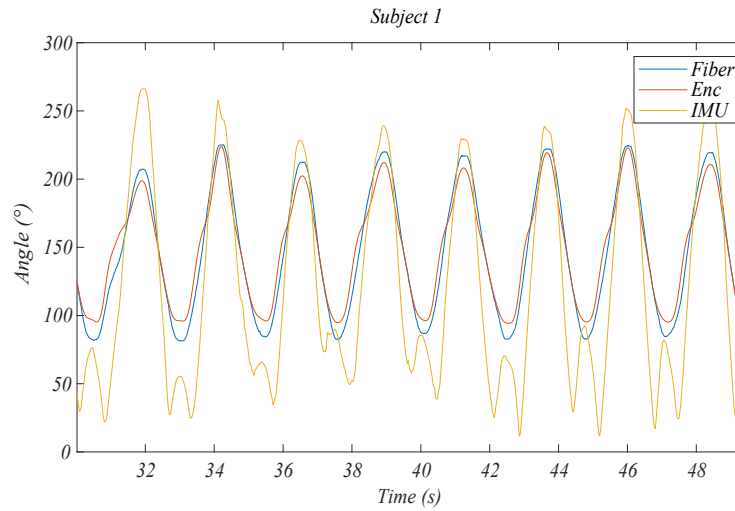


Figure 3. Optical fiber characterization for flexion and extension of the elbow

Table 2. Mean and standard deviation of variables of 5 subjects performing elbow flexion/extension. Fiber and IMU errors are presented. (*) indicate the normal distribution of data.

Parameter/Sensor	Encoder	Fiber	Fiber error (%)	IMU	IMU error (%)	p-value
ROM	139.0 $\pm$ 6.9	121.7 $\pm$ 9.9	12.44	220.2 $\pm$ 45.8*	58.39	p<0.001
RMS	155.4 $\pm$ 3.2	154.8 $\pm$ 3.0	-	145.3 $\pm$ 18.0*	6.53	p<0.001
MEAN	148.1 $\pm$ 3.4	149.7 $\pm$ 2.9	-	125.8 $\pm$ 13.9*	14.97	p<0.001
MIN	83.1 $\pm$ 2.0*	95.1 $\pm$ 0.7	14.44	21.8 $\pm$ 16.6	73.75	p<0.001
MAX	222.3 $\pm$ 4.5	217.5 $\pm$ 7.8	2.16	252.4 $\pm$ 33.3	13.54	p<0.001

serve that the OF sensor, compared to the magnetic encoder, is a low-cost sensor that can be used in rigid and soft robotic devices due to its flexible characteristics [4], however, it has a limitation related to variable losses which leads to low sensitivity and could cause errors when compared to the magnetic encoder. Furthermore, it is immune to magnetic fields, an issue commonly present in magnetic encoders in robotics applications [1]. This feature provides greater reliability and stability in the measurements, reducing the potential for false or inaccurate readings. The flexibility and immunity to electromagnetic interference of the OF sensor make it a suitable option for soft robotics applications.

3. Conclusion

In conclusion, the results of this study provide valuable information for selecting a suitable measurement device for exoskeleton instrumentation. The OF sensor proved to have multiple advantages over the IMU sensor obtaining errors between 2% - 14% as compared to errors between 6% -73% respectively. It is also low-cost, flexible, and immune to electromagnetic interference, making it an affordable option to measure angle values for robotic devices.

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